

```
In [15]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [16]: df = pd.read_csv('EDA Using Python.csv')
df
```

Out[16]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService
0	7590-VHVEG	Female	0	Yes	No	1	No
1	5575-GNVDE	Male	0	No	No	34	Yes
2	3668-QPYBK	Male	0	No	No	2	Yes
3	7795-CFOCW	Male	0	No	No	45	No
4	9237-HQITU	Female	0	No	No	2	Yes
...	...	...	...	...	...	...	...
7038	6840-RESVB	Male	0	Yes	Yes	24	Yes
7039	2234-XADUH	Female	0	Yes	Yes	72	Yes
7040	4801-JJAZL	Female	0	Yes	Yes	11	No
7041	8361-LTMKD	Male	1	Yes	No	4	Yes
7042	3186-AJIEK	Male	0	No	No	66	Yes

7043 rows × 21 columns



```
In [16]: df.head()
```

Out[16]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	Mul
--	------------	--------	---------------	---------	------------	--------	--------------	-----

0	7590-VHVEG	Female	0	Yes	No	1	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	

5 rows × 21 columns



In [17]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                 7043 non-null   object
2   SeniorCitizen          7043 non-null   int64
3   Partner                7043 non-null   object
4   Dependents             7043 non-null   object
5   tenure                 7043 non-null   int64
6   PhoneService           7043 non-null   object
7   MultipleLines          7043 non-null   object
8   InternetService        7043 non-null   object
9   OnlineSecurity         7043 non-null   object
10  OnlineBackup           7043 non-null   object
11  DeviceProtection       7043 non-null   object
12  TechSupport            7043 non-null   object
13  StreamingTV            7043 non-null   object
14  StreamingMovies        7043 non-null   object
15  Contract               7043 non-null   object
16  PaperlessBilling       7043 non-null   object
17  PaymentMethod          7043 non-null   object
18  MonthlyCharges         7043 non-null   float64
19  TotalCharges           7043 non-null   object
20  Churn                  7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

In [18]: `df['TotalCharges'] = df["TotalCharges"].replace(" ", "0")`
`df['TotalCharges'] = df["TotalCharges"].astype("float")`

In [19]: `df.info()`

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                 7043 non-null   object
2   SeniorCitizen          7043 non-null   int64
3   Partner                7043 non-null   object
4   Dependents             7043 non-null   object
5   tenure                 7043 non-null   int64
6   PhoneService           7043 non-null   object
7   MultipleLines          7043 non-null   object
8   InternetService        7043 non-null   object
9   OnlineSecurity         7043 non-null   object
10  OnlineBackup           7043 non-null   object
11  DeviceProtection       7043 non-null   object
12  TechSupport            7043 non-null   object
13  StreamingTV            7043 non-null   object
14  StreamingMovies        7043 non-null   object
15  Contract               7043 non-null   object
16  PaperlessBilling       7043 non-null   object
17  PaymentMethod          7043 non-null   object
18  MonthlyCharges         7043 non-null   float64
19  TotalCharges           7043 non-null   float64
20  Churn                  7043 non-null   object
dtypes: float64(2), int64(2), object(17)
memory usage: 1.1+ MB

```

in that we replace blank space with 0

In [20]: `df.isnull()`

Out[20]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService
0	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...
7038	False	False	False	False	False	False	False
7039	False	False	False	False	False	False	False
7040	False	False	False	False	False	False	False
7041	False	False	False	False	False	False	False
7042	False	False	False	False	False	False	False

7043 rows × 21 columns

In [21]: `df.isnull().sum()`

```
Out[21]: customerID      0
gender      0
SeniorCitizen  0
Partner      0
Dependents    0
tenure      0
PhoneService  0
MultipleLines  0
InternetService  0
OnlineSecurity  0
OnlineBackup  0
DeviceProtection  0
TechSupport    0
StreamingTV    0
StreamingMovies  0
Contract      0
PaperlessBilling  0
PaymentMethod  0
MonthlyCharges  0
TotalCharges  0
Churn         0
dtype: int64
```

In [22]: `df.isnull().sum().sum()`

Out[22]: `np.int64(0)`

In [23]: `df.describe()`

```
Out[23]:
```

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2279.734304
std	0.368612	24.559481	30.090047	2266.794470
min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	398.550000
50%	0.000000	29.000000	70.350000	1394.550000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

In [24]: `df.duplicated()`

```
Out[24]: 0      False
         1      False
         2      False
         3      False
         4      False
         ...
        7038    False
        7039    False
        7040    False
        7041    False
        7042    False
        Length: 7043, dtype: bool
```

```
In [25]: df.duplicated().sum()
```

```
Out[25]: np.int64(0)
```

```
In [26]: df['customerID'].duplicated().sum()
```

```
Out[26]: np.int64(0)
```

```
In [27]: def conv(value):
         if value == 1:
             return "yes"
         else:
             return "no"
         df['SeniorCitizen']=df["SeniorCitizen"].apply(conv)
```

convert 0 and 1 to yes and no to make it easier to analyze

```
In [28]: df.head(30)
```

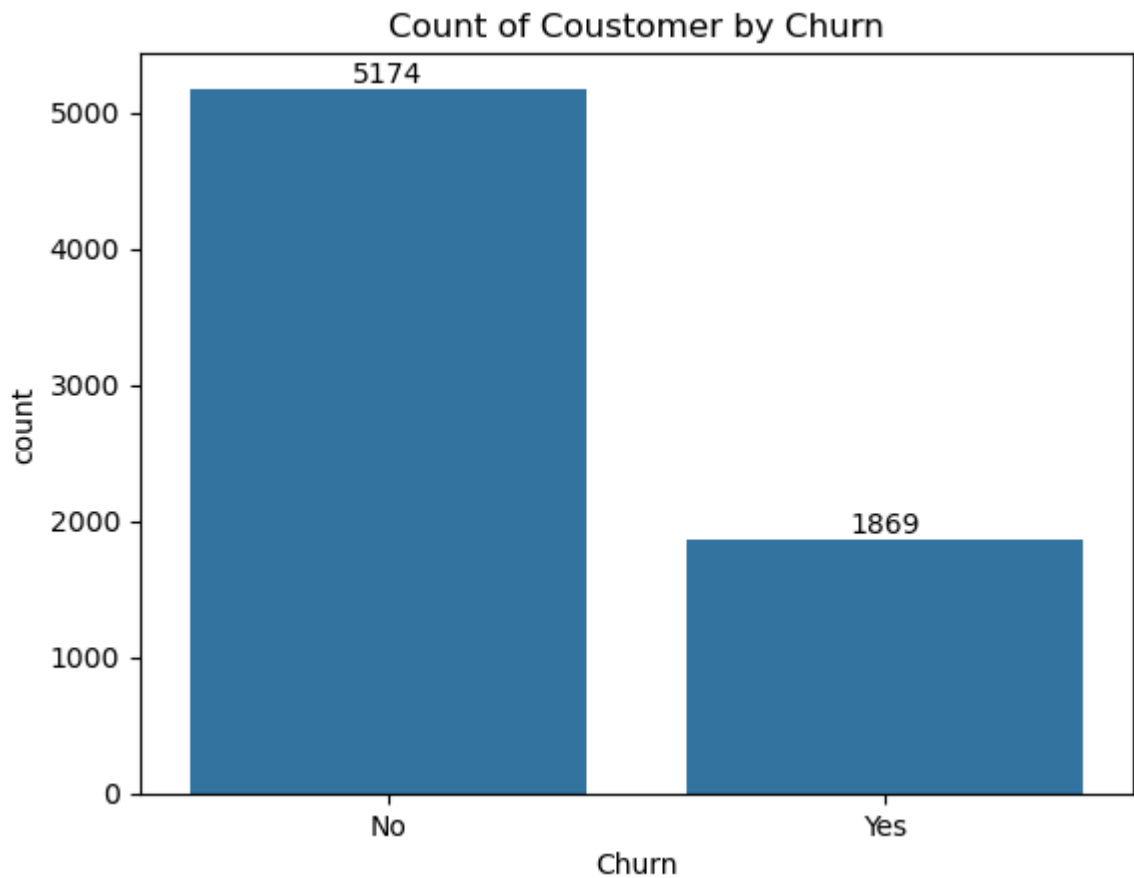
Out[28]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	Mi
0	7590-VHVEG	Female	no	Yes	No	1	No	
1	5575-GNVDE	Male	no	No	No	34	Yes	
2	3668-QPYBK	Male	no	No	No	2	Yes	
3	7795-CFOCW	Male	no	No	No	45	No	
4	9237-HQITU	Female	no	No	No	2	Yes	
5	9305-CDSKC	Female	no	No	No	8	Yes	
6	1452-KIOVK	Male	no	No	Yes	22	Yes	
7	6713-OKOMC	Female	no	No	No	10	No	
8	7892-POOKP	Female	no	Yes	No	28	Yes	
9	6388-TABGU	Male	no	No	Yes	62	Yes	
10	9763-GRSKD	Male	no	Yes	Yes	13	Yes	
11	7469-LKBCI	Male	no	No	No	16	Yes	
12	8091-TTVAX	Male	no	Yes	No	58	Yes	
13	0280-XJGEX	Male	no	No	No	49	Yes	
14	5129-JLPIS	Male	no	No	No	25	Yes	
15	3655-SNQYZ	Female	no	Yes	Yes	69	Yes	
16	8191-XWSZG	Female	no	No	No	52	Yes	

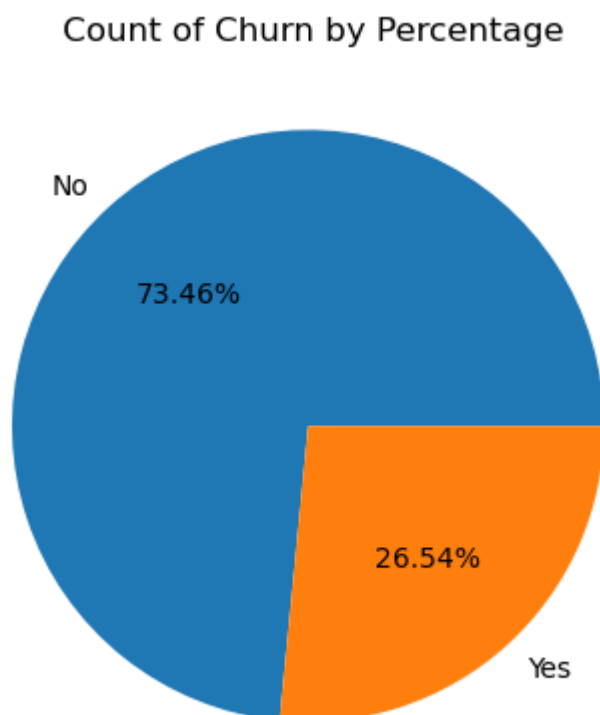
	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	Mi
17	9959-WOFKT	Male	no	No	Yes	71	Yes	
18	4190-MFLUW	Female	no	Yes	Yes	10	Yes	
19	4183-MYFRB	Female	no	No	No	21	Yes	
20	8779-QRDMV	Male	yes	No	No	1	No	
21	1680-VDCWW	Male	no	Yes	No	12	Yes	
22	1066-JKSGK	Male	no	No	No	1	Yes	
23	3638-WEABW	Female	no	Yes	No	58	Yes	
24	6322-HRPFA	Male	no	Yes	Yes	49	Yes	
25	6865-JZDKO	Female	no	No	No	30	Yes	
26	6467-CHFZW	Male	no	Yes	Yes	47	Yes	
27	8665-UTDZH	Male	no	Yes	Yes	1	No	
28	5248-YGIJN	Male	no	Yes	No	72	Yes	
29	8773-HHUOZ	Female	no	No	Yes	17	Yes	

30 rows × 21 columns

```
In [29]: ax=sns.countplot(x='Churn',data=df)
ax.bar_label(ax.containers[0])
plt.title("Count of Coustomer by Churn")
plt.show()
```



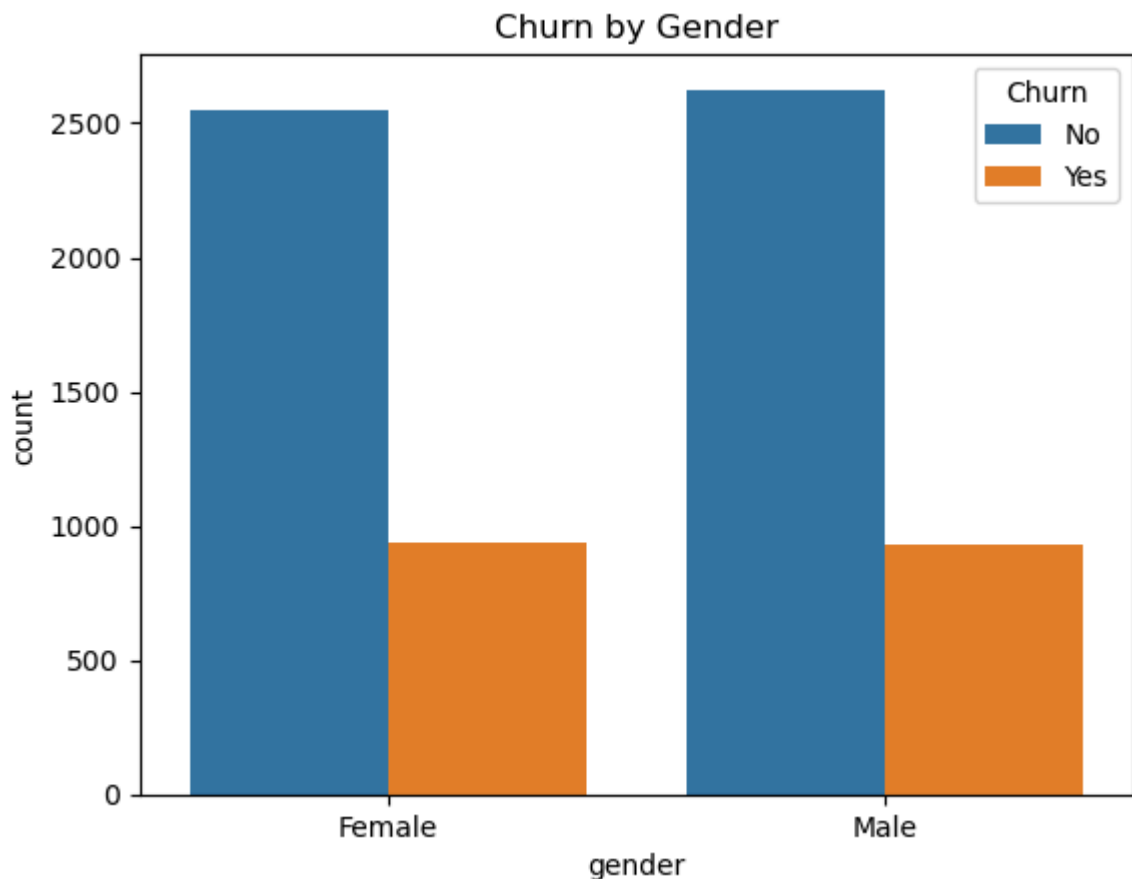
```
In [30]: gb = df.groupby("Churn").agg({'Churn':"count"})  
plt.pie(gb['Churn'], labels = gb.index, autopct = "%1.2f%%")  
plt.title(" Count of Churn by Percentage")  
plt.show()
```



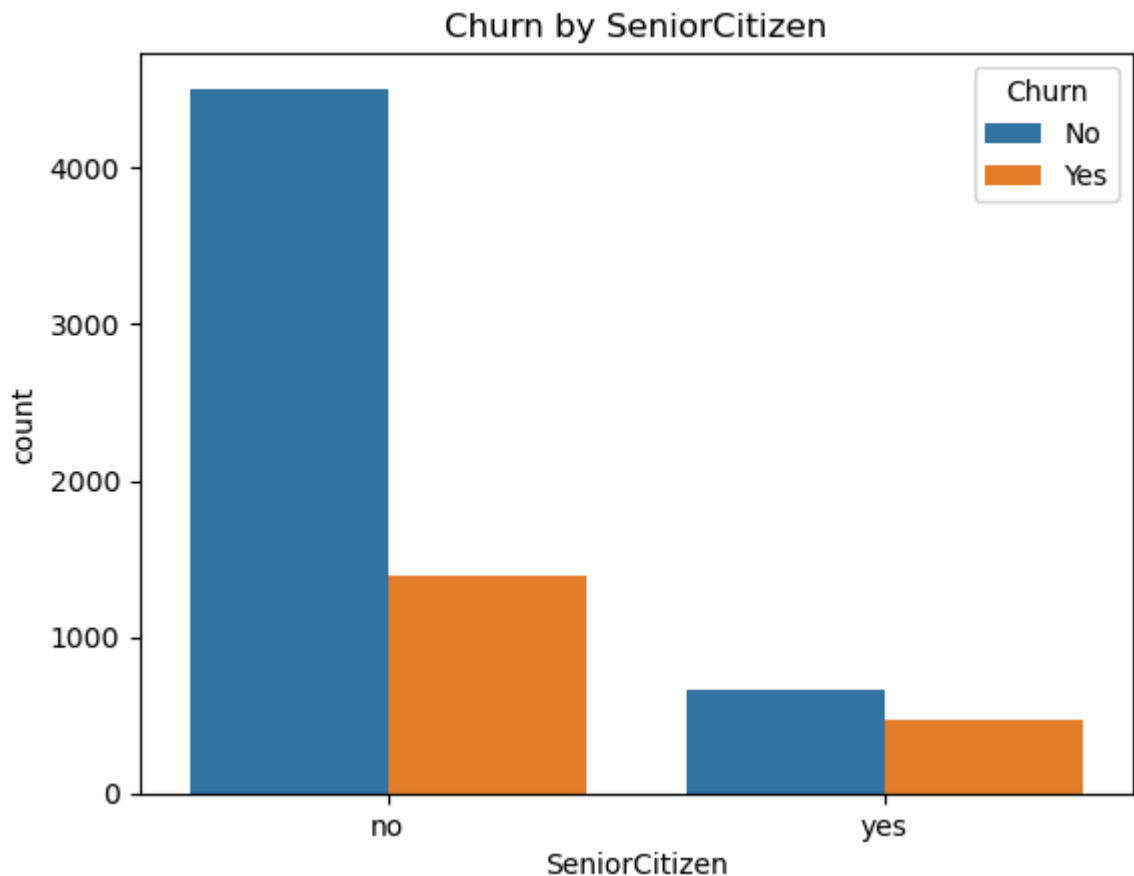
From the pie chart we able to say that 26.54 % of people are Churned out

Now Explore the reason behind it

```
In [31]: sns.countplot(x = "gender", data = df, hue = 'Churn')  
plt.title('Churn by Gender')  
plt.show()
```



```
In [32]: sns.countplot(x = "SeniorCitizen", data = df, hue = 'Churn')  
plt.title('Churn by SeniorCitizen')  
plt.show()
```

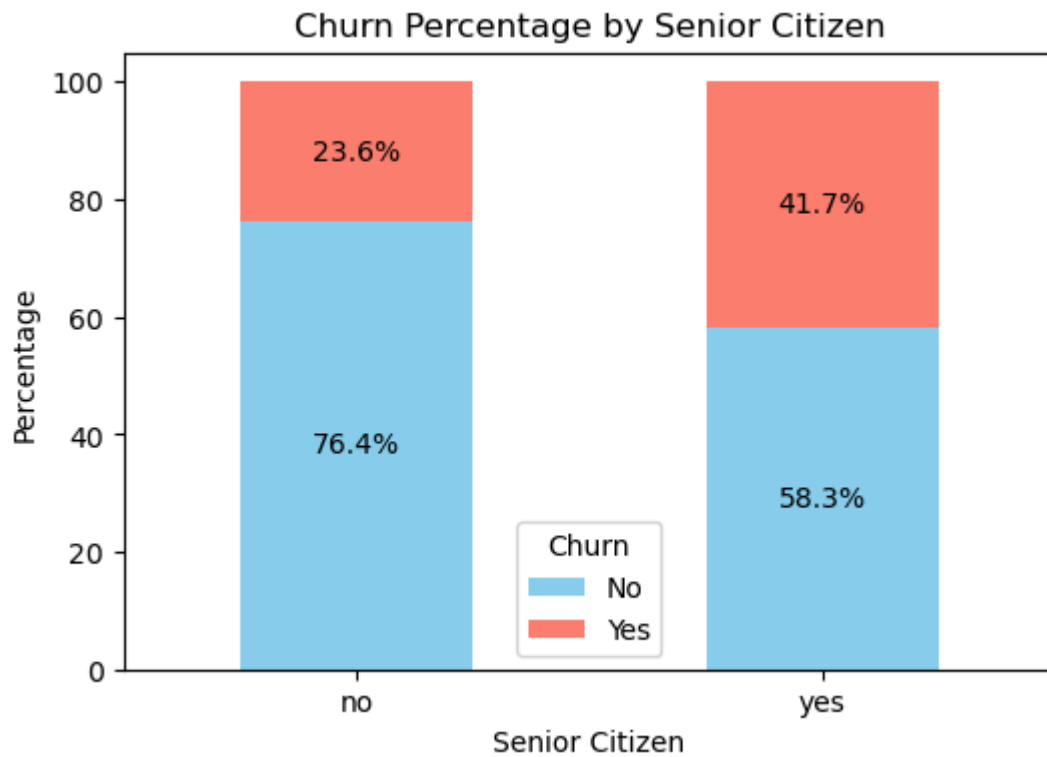


```
In [33]: # Percentage table
churn_pct = pd.crosstab(
    df['SeniorCitizen'],
    df['Churn'],
    normalize='index'
) * 100

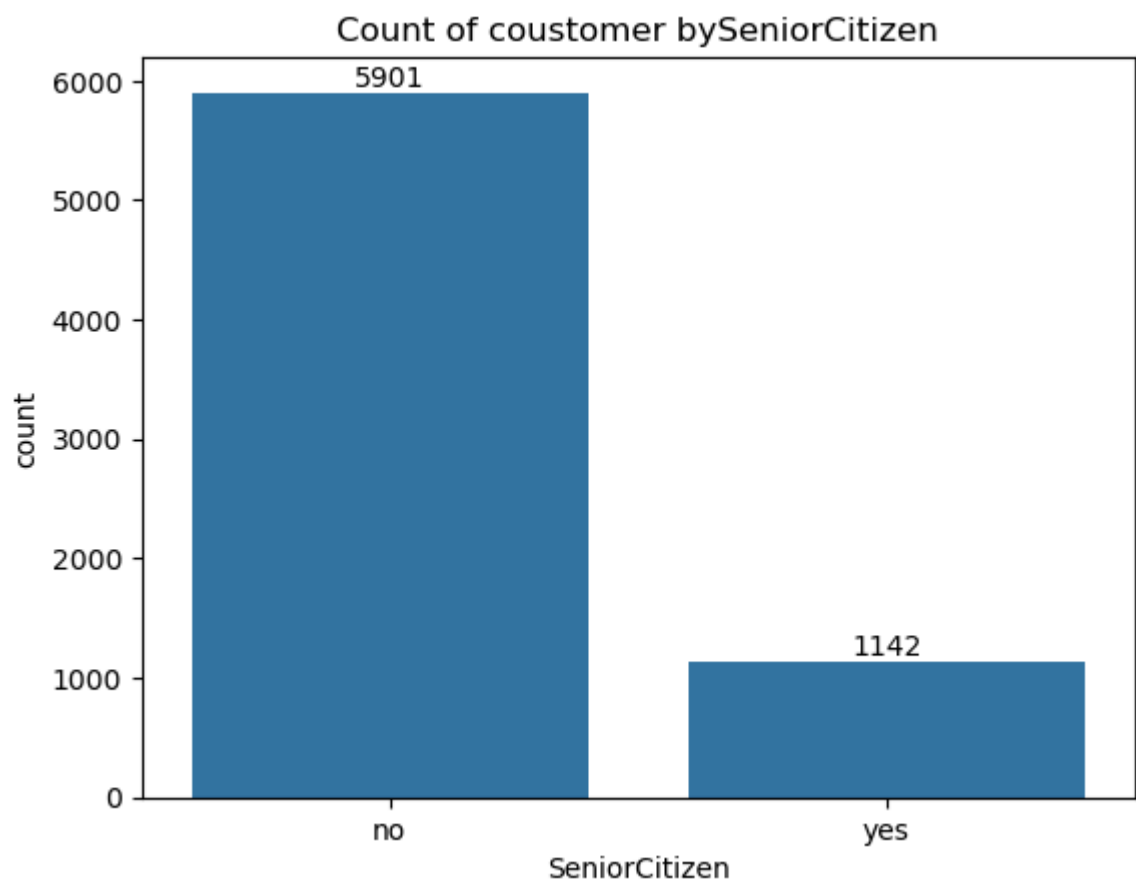
ax = churn_pct.plot(
    kind='bar',
    stacked=True,
    figsize=(6,4),
    color=['skyblue', 'salmon']
)

# Add percentage Labels
for container in ax.containers:
    labels = [f'{v:.1f}%' if v > 0 else '' for v in container.datavalues]
    ax.bar_label(container, labels=labels, label_type='center', color='black')

plt.title('Churn Percentage by Senior Citizen')
plt.xlabel('Senior Citizen')
plt.ylabel('Percentage')
plt.legend(title='Churn')
plt.xticks(rotation=0)
plt.show()
```

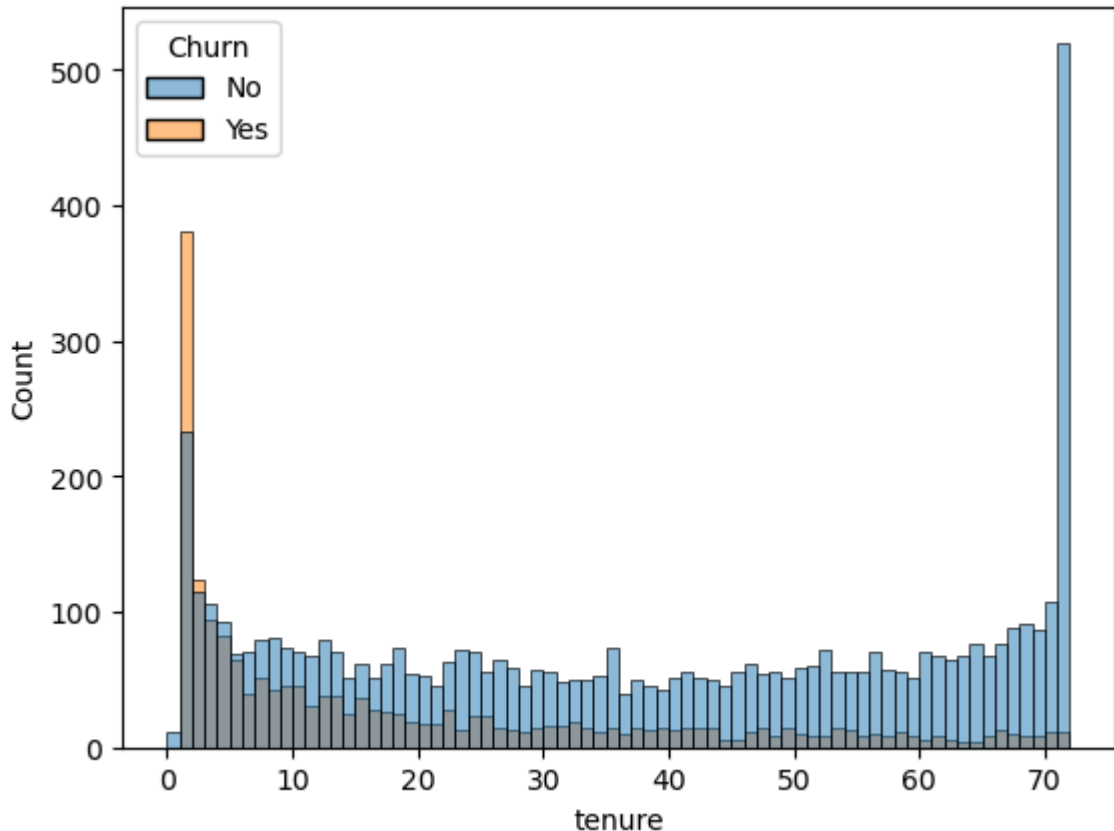


```
In [34]: ax = sns.countplot(x = "SeniorCitizen", data = df)
ax.bar_label(ax.containers[0])
plt.title('Count of coustomer bySeniorCitizen')
plt.show()
```



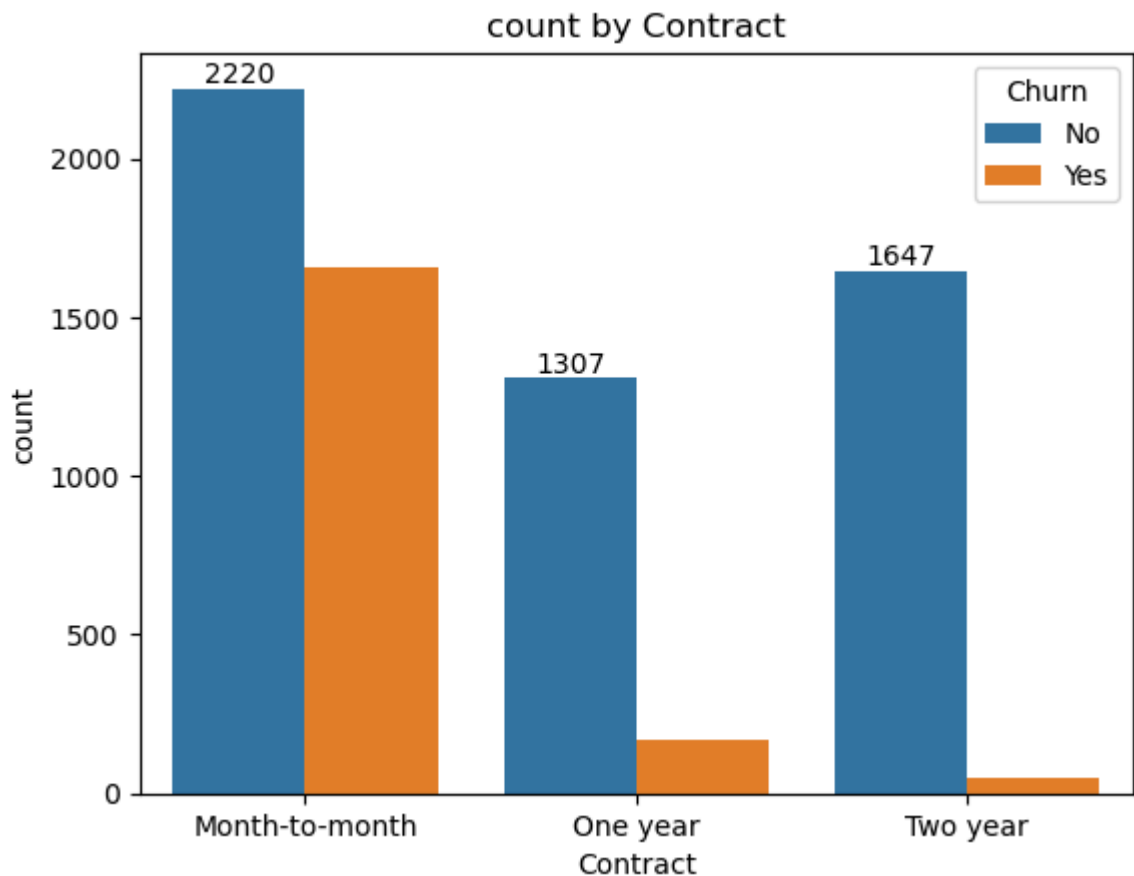
comparitively a large amount of data is churned out in senior citizen

```
In [35]: sns.histplot(x="tenure", data=df, bins = 72, hue= "Churn")
plt.show()
```



people who stay for a month or two month those churn out more compare to those who stay for long it stay

```
In [36]: ax= sns.countplot(x = "Contract", data = df, hue = 'Churn')
ax.bar_label(ax.containers[0])
plt.title('count by Contract')
plt.show()
```



people who have monthly plan the churn more compare to those who have yearly plan

In [37]: `df.columns.values`

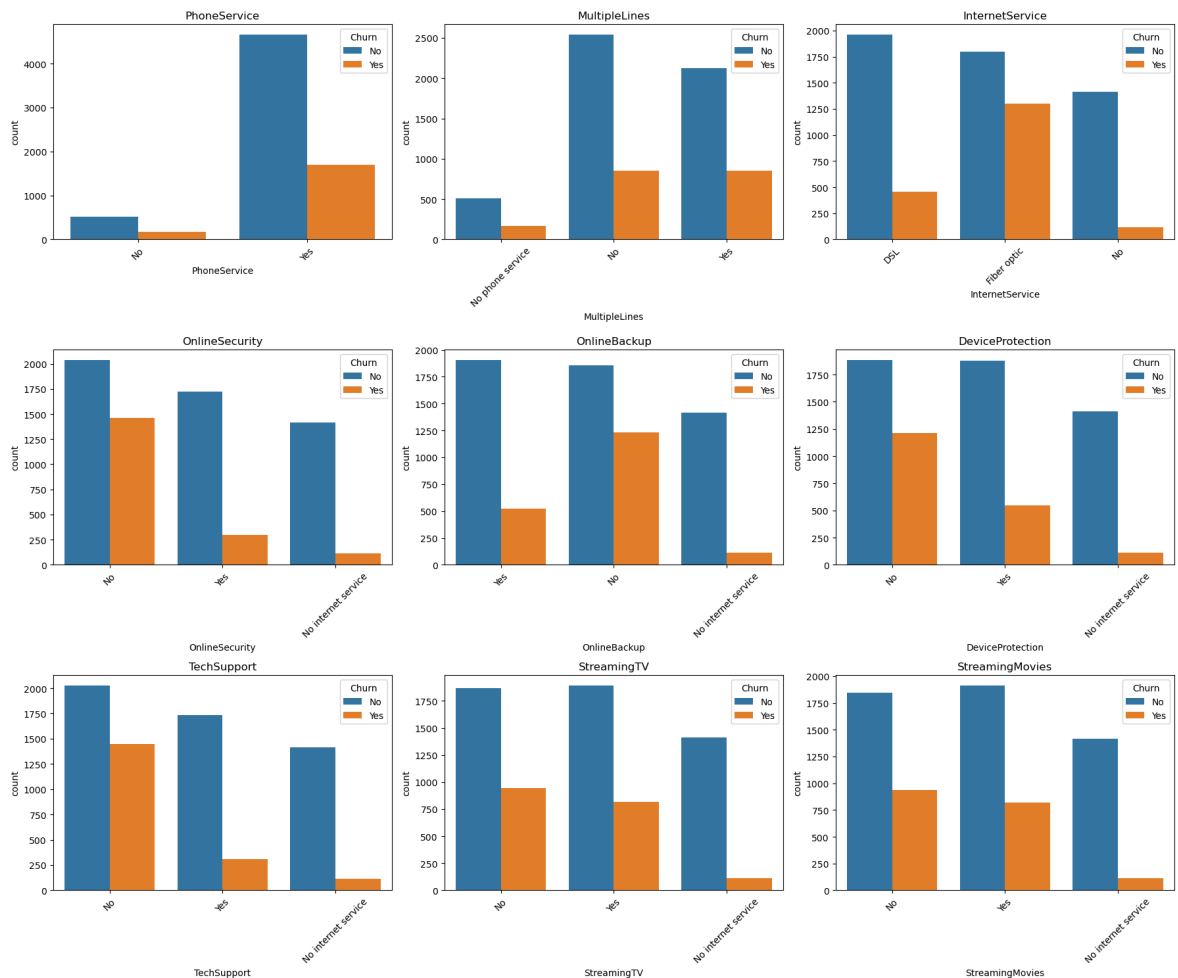
Out[37]: `array(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'], dtype=object)`

```
In [39]: cols = [
    'PhoneService', 'MultipleLines', 'InternetService',
    'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
    'TechSupport', 'StreamingTV', 'StreamingMovies'
]

# Create a 3x3 grid of subplots
fig, axes = plt.subplots(3, 3, figsize=(18, 15))
axes = axes.flatten()

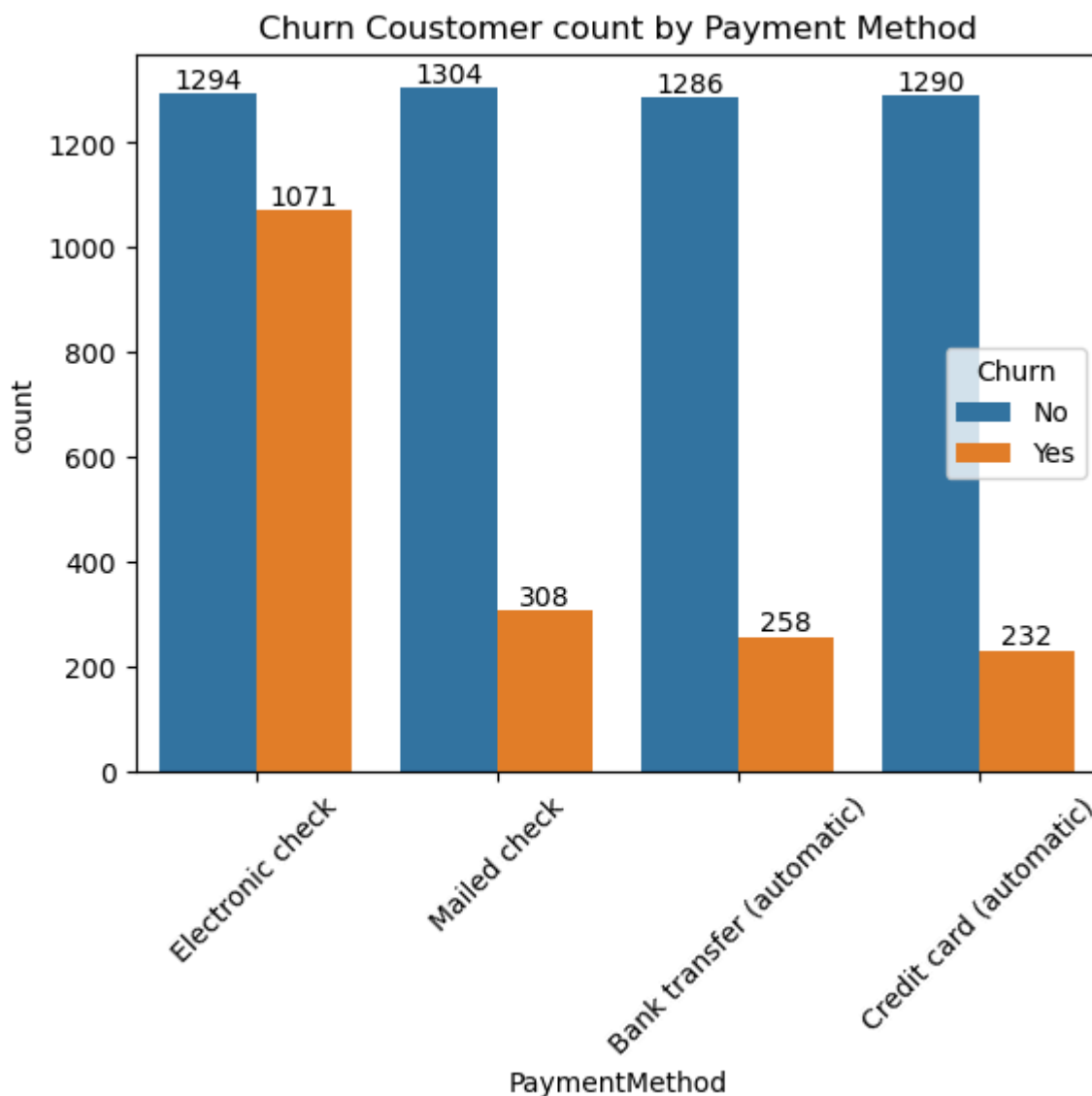
for i, col in enumerate(cols):
    sns.countplot(data=df, x=col, ax=axes[i], hue = df["Churn"])
    axes[i].set_title(col)
    axes[i].tick_params(axis='x', rotation=45)
```

```
plt.tight_layout()
plt.show()
```



Customers using **Fiber Optic internet** exhibit the highest churn rates, while **DSL users** show comparatively better retention, indicating internet service type is a strong predictor of churn. * Customers who **do not subscribe to Online Security, Online Backup, Device Protection, or Tech Support** are significantly more likely to churn than those who have these services. * Churn differences for **Streaming TV** and **Streaming Movies** are less pronounced, suggesting entertainment services have a weaker impact on customer retention. * Overall, customers with additional support and protection services tend to remain with the company longer, highlighting the importance of bundled value-added services in reducing churn.

```
In [45]: ax= sns.countplot(x = "PaymentMethod",data = df, hue = 'Churn')
ax.bar_label(ax.containers[0])
ax.bar_label(ax.containers[1])
plt.title('Churn Coustomer count by Payment Method')
plt.xticks(rotation = 45)
plt.show()
```



coustomer is likely to churn when he uses payment Method

In []: